

Networking and communication:

Networking and communication components enable real-time interactions between users within the metaverse. They handle the transmission of data, such as avatar movements, chat messages, and collaborative activities, ensuring a seamless and responsive virtual experience.

Content creation and management:

The content creation and management component provides tools and interfaces for users and developers to create and manage content within the metaverse. This includes creating 3D models, textures, animations, and scripts, as well as managing user-generated content.

Collaboration tools

Collaboration tools foster community engagement and teamwork within the metaverse. They include features such as shared workspaces, project collaboration tools, and content sharing platforms that enable users to work together on creative projects and shared experiences.

Social interaction: Social interaction components facilitate communication and socialisation among users. This may include text chat, voice chat, and emotes, allowing users to connect and interact with each other, making the metaverse a social and collaborative space.

API and integration: API (application programming interface) and integration components allow developers to extend the functionalities of the open source metaverse platform. They enable integration with external services, plugins, and third-party applications, fostering innovation and flexibility within the platform.

Data management: Data management to handle digital assets, templates for virtual platforms, data protection against user data and customised user experience related data are handled with these services to facilitate efficient data-driven user experience design.

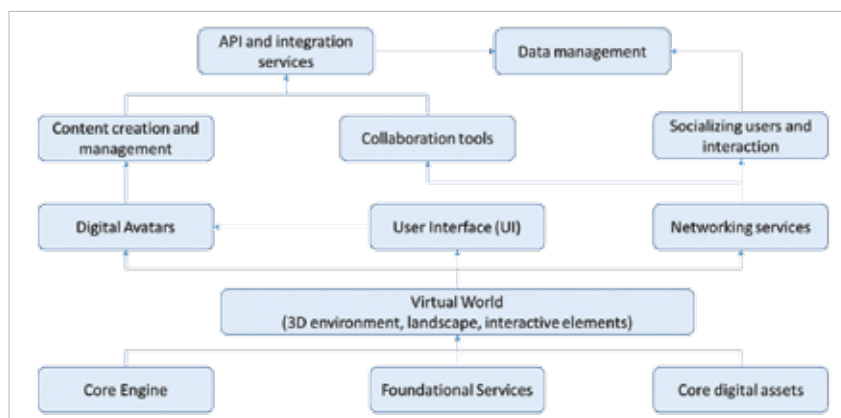


Figure 1: Key components of open source metaverse platforms

Advantages of open source metaverse platforms

Open source metaverse platforms offer several advantages that contribute to their popularity and potential.

Flexibility and customisation:

These platforms allow users to modify and customise the software according to their specific needs and preferences. This flexibility enables developers and creators to build unique virtual experiences tailored to their target audience.

Community collaboration: By making the metaverse platform open source, developers from around the world can contribute to its development, identify and fix bugs, and add new features. This collaborative approach fosters innovation and accelerates the platform's growth.

Lower costs: Open source metaverse platforms eliminate the need for expensive licensing fees. Individuals and organisations can use and modify the software without incurring significant costs, making it more accessible to a wider range of users. This affordability factor promotes inclusivity and encourages diverse participation.

Enhanced security and privacy:

With open source platforms, the source code is openly available for inspection and review by a global community. This transparent approach can enhance security by allowing experts to identify and fix vulnerabilities promptly. Open source projects also tend to prioritise

privacy and data protection, giving users greater control over their virtual identities and personal information.

Interoperability and standards:

Open source metaverse platforms often focus on interoperability, enabling users to seamlessly move between different virtual environments. Standardisation efforts within the open source community facilitate the integration of various components, tools, and technologies, creating a more interconnected metaverse experience.

Avoiding vendor lock-in:

These platforms provide users with independence from a single vendor or company. By avoiding vendor lock-in, users have the freedom to switch between different platforms or even create their own versions, reducing dependence on a specific provider and promoting healthy competition.

Educational and research opportunities:

Open source metaverse platforms offer valuable educational and research opportunities. Students, researchers, and developers can study the code base, experiment with new ideas, and contribute to the advancement of virtual reality technologies.

Innovation and experimentation:

Open source metaverse platforms often act as incubators for innovation and experimentation. The accessibility of the code base encourages developers to push boundaries, create new features, and explore novel use cases. This vibrant